CHAPTER

12

Principles of Cavity Preparation

The most prevalent and the most common ailment of oral cavity, the caries, is frequently confronted by all dental professionals. Removal of caries and preparing proper foundation (cavity preparation) for the restorative material is the key to success for any clinician. It is mandatory for the operator to be thoroughly apprised with all the fundamentals of cavity preparation.

A cavity in dentistry is defined as "*a defect in enamel, dentin or cementum resulting from the pathological processes, mostly the dental caries*". Other processes such as abrasion, erosion, etc. can also cause such defects.

Preparation of the cavity includes "*the performance of all mechanical procedures required to remove the carious and affected tissues and to shape the remaining tooth tissues so as to receive a biologically and mechanically sound restoration*".

The preparation of a cavity should be carried out in an orderly sequence, following certain principles. Dr GV Black suggested six principles for cavity preparation, which by and large are followed; though each principle has been modified keeping in view the advancement in restorative materials. The principles of cavity preparation are as follows:

1. Outline form
2. Resistance and retention forms
3. Convenience form
4. Removal of remaining carious dentin
5. Finishing the enamel walls and margins
6. Toilet of the cavity

The cavity preparation is accomplished following systematic procedures. Thorough knowledge of tooth anatomy is mandatory to understand these principles. The direction of enamel rods, thickness of enamel-dentin at various places, position of the pulp and the relation of crown with the gingival tissue must be kept in mind to facilitate proper cavity preparation. The physical and mechanical properties of the restorative materials to be used should also be taken care of during cavity preparation.

1. OUTLINE FORM

Dr GV Black described outline form as "*the area of tooth surface or the enamel margin to be included in the finished cavity*".

This is basically the external outline form (implies placing the margins of the cavity to the position it will be at the final preparation). The internal outline form, however, includes the inner dimensions (initial depth) of the prepared cavity.

One should visualize the outline form in one's mind before any tooth (cavity) preparation is started. It prevents over-cutting and over-extension, which often leaves weak remaining tooth structure.

General features influencing outline form are:

- Location of the carious lesion
- Extent of the carious lesion
- Position of pits and fissures
- Proximity of the lesion to other defects in the enamel
- Relationship of adjacent and opposing teeth
- Relationship of soft tissue
- Type of restorative material to be used
- Functional requirements of the restoration
- Esthetic considerations

Establishing Outline Form

a. *Occlusal Cavities (Class I)*

- Extend the cavity margins to sound tooth structure
- Remove all unsupported enamel (weak enamel margins).
- Include all susceptible fissures in the outline form (extension for prevention).
  (The phenomenon of 'extension for prevention' was first suggested by Marshall Ebb and was later adopted by Black).
- When two cavities have less than 1.0 mm of sound tooth structure between them, they should be joined.
- Cavity margins should not terminate in high stress areas, such as cusp heights or ridge crests.
- Extend the margins to allow sufficient access for proper cavity preparation, restoration placement and finishing procedures.

- The depth of cavity preparation should be 0.2–0.5 mm into dentin (below dentinoenamel junction); total depth may be 1.5–2.0 mm (varies because of cusp heights).
- In case, where the decay in dentin has undermined enamel, especially under the cusps and up to line angles of the tooth, the removal of caries along with unsupported enamel is mandatory. This procedure might necessitate cuspal coverage or full veneer restoration.
- In occlusal outline form, the buccolingual intercuspal extension, if greater than two-thirds of the cusp incline, reduces the strength of the cusps; less than 1.0 mm, leads to poor condensation, resulting in porosity and poor adaptation of the restoration.
- Figure 12.1A to C shows the outline form for occlusal cavities (Class I).

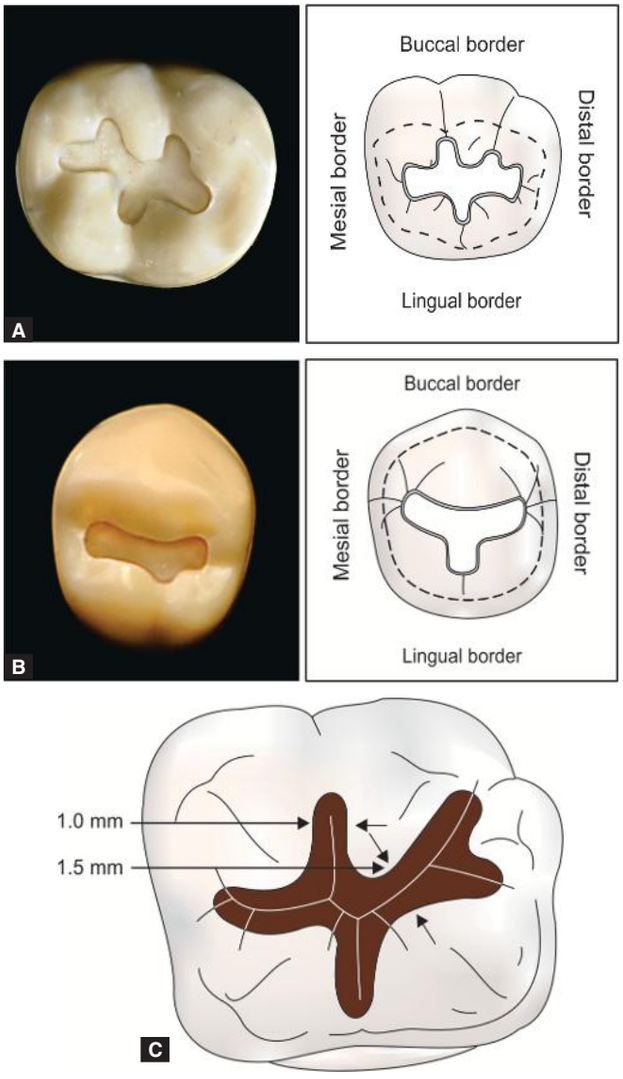


Fig. 12.1: Occlusal outline form (Class I): **(A)** Mandibular first molar; **(B)** Mandibular second premolar; **(C)** Occlusal dimensions

b. *Proximal Cavities (Class II and III)*

- The preparation of occlusal area is as governed for occlusal cavities.
- Position and crest of healthy gingiva, age of the patient vis-a-vis the epithelial attachment and the scope for future gingival recession.
- Dimensions of contact area (contact area is to be involved and later restored to same configuration).
- Extent of caries at and around the contact area.
- Oral hygiene of the patient.
- Possible forces of mastication, which the restoration will be withstanding.
- Axial margins should be in self-cleansing areas (Fig. 12.2).
- The depth of axial wall should be 0.5–0.8 mm into dentin (inside dentinoenamel junction).
- Axial wall should follow the outer contour of the tooth surface (practically parallel to external tooth surface).
- Extend the cavity margins to sound tooth structure.
- All unsupported enamel rods (weak enamel margins) should be removed.
- In proximal cavities of posterior teeth (Class II), the gingival seat is placed apical to contact area (should have 0.5 mm clearance from the adjacent tooth).
- In proximal outline form, the buccolingual extension should not be less than 1.5 mm and the occlusogingival extension should not be less than 3.0 mm. (*Occlusogingival extension is measured relative to the marginal ridge; usual mean distance from the marginal ridge to the cementoenamel junction is 5.0 mm for premolars and 6.0 mm for molars.)*
- In proximal cavities of anterior teeth (Class III), the incisal extension is kept up to the incisal margin.

Figure 12.3A to C shows the outline form for proximal cavity in posterior tooth and Fig. 12.4A to C shows the outline form for proximal cavity in anterior tooth.

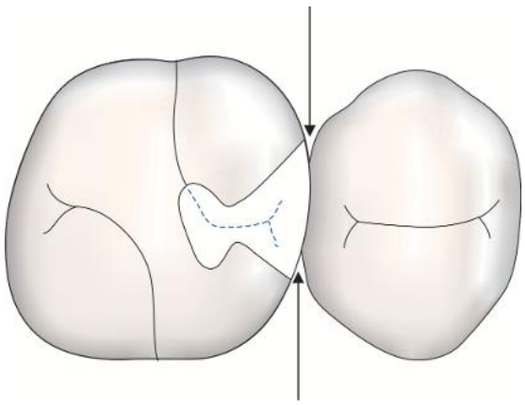


Fig. 12.2: Axial margins clearing the contact area and extend into embrasures (arrows)

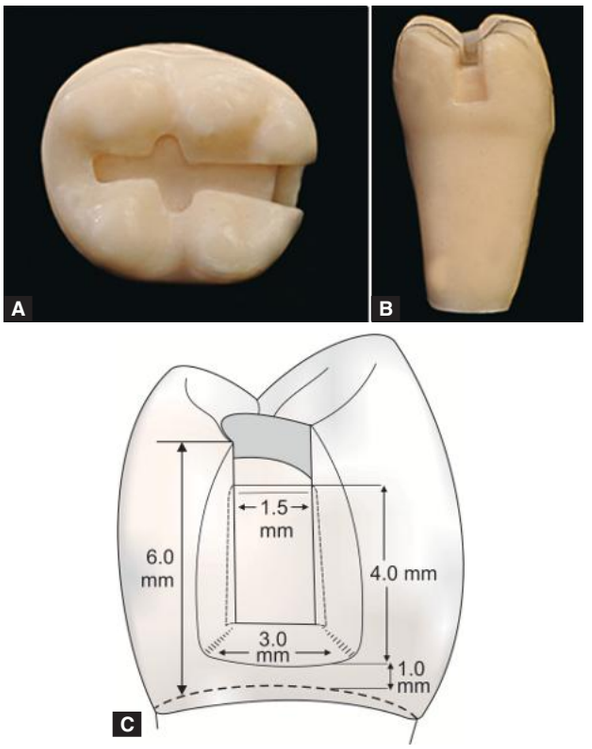


Fig. 12.3: Outline form for proximal cavity in mandibular first molar (Class II): **(A)** Occlusal view; **(B)** Proximal view; **(C)** Proximal dimensions

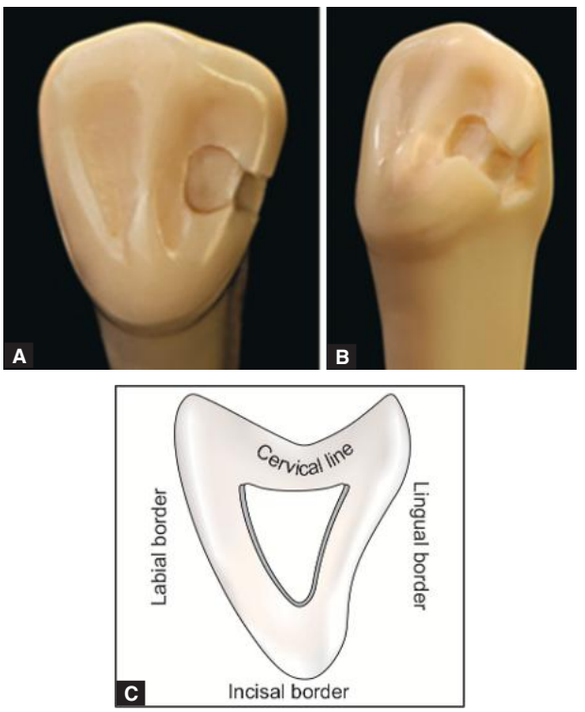


Fig. 12.4: Outline form for proximal cavity in maxillary canine (Class III): **(A)** Palatal view; **(B)** Mesiopalatal view; **(C)** Diagrammatic (mesial aspect)

c. Gingival/Cervical Cavities (Class V)

- The cavity is prepared till sound tooth structure is achieved in all directions, viz. occlusally, gingivally, mesially and distally.
- Health and position of the gingiva: In young patients, the restoration is covered by gingiva, i.e. most of the restorative margins in such patients are kept subgingival. However, in older patients, with the recession, the margins can be kept supragingival.
- Forces of mastication, abnormal occlusal contacts and eating pattern of the patient should also be kept in mind while planning cervical cavities.
- Contour of the buccal/lingual surfaces
- Oral hygiene of the patient, vis-à-vis brushing habits, and the quality of brush.
- Mesial and distal extensions should be placed in self-cleansing areas.
- Depth of axial wall should be 0.2–0.5 mm inside dentinoenamel junction. (In cemental caries, the depth into dentin is kept less.)
- The axial wall should follow the outer contour of the tooth surface (curved on buccal sides and approximately straight on lingual sides).

Figure 12.5 shows the outline form for Class V cavity in central incisor and maxillary molar.

The outline form for root caries is controlled mainly by the site of the lesion, extent of the caries, age of the patient and the oral hygiene. By and large, root caries is managed by excavating the caries and restoring without any injudicious extension.

Modifications of Outline Form

The basic principles can be modified; however, these modifications should be within limits so as the remaining tooth structure be preserved as far as possible.

Conditions which warrant reduced extension of cavity margins are:

- Proximal cavities (for newer materials; conservative cavity preparation)
- Root proximities
- Esthetic requirements

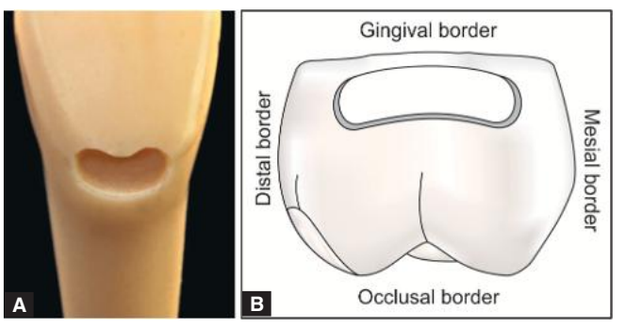


Fig. 12.5: **(A)** Outline form for cervical cavity in maxillary central incisor (Class V); **(B)** Diagrammatic (maxillary molar)

Conditions which warrant increased extension of cavity margins are:

- Mental or physical handicap patients
- Advanced age of the patient
- When additional resistance or retention is required
- Need to adjust tooth contours

Various authors have questioned the principle of 'extension for prevention'. *[(i) All pits and fissures are involved (may and may not be carious); the cavity margins be placed in non-fissured areas, (ii) Proximal line angle extend buccally and lingually in 'self-cleansing areas', (iii) The cervical margins extend below the gingival margin.]*

It has been established that the gingival sulcus is not bacteria free and the self-cleansing areas are virtually non-existent. Therefore, there is judicious need to extend the cavity margins in self cleansing areas. In Class II amalgam preparation, the objective of preparation of the occlusal portion is to provide access to the proximal lesion and to remove defective pits and fissures. Various studies have concluded that the incidence of isthmus fracture is reduced in conservative cavity preparations.

The concept of '*extension for prevention*' is visualized as '*prevention of extension*' in the present day restorative dentistry. The advancement of restorative materials in the form of chemical bonding (bonding agents) and also remineralizing agents (calcium phosphate, fluorides, etc.) have questioned the validity of extension for prevention.

The external and internal outline forms are to be modified as per requirements of newer restorative materials.

The outline form for composite restorations should be shallow and narrow as compared to amalgam. The low tensile strength of amalgam dictates bulk placement, especially where heavy forces are applied. Amalgam requires a minimal cavity depth of 2.0 mm with an adequately wide isthmus. Composites, because of their high tensile strength, permit a narrow and shallow isthmus. As composites undergo volumetric shrinkage of 1.5 to 2.0% during polymerization, a minimal cavity outline reduces the tendency for open margins. The proximal outline is modified; the cavosurface margins are not placed in embrasure areas (placed within the contact area).

In composites, the enamel margins are beveled to promote better sealing. Enamel in the proximal wall should be beveled because prism direction is at right angle to the cavosurface. Occlusal bevels are unnecessary because the prism directions in the zone of central fossa are inclined towards the fossa. It has been shown that non-beveled occlusal surfaces provided better sealing against microleakage in comparison with a beveled margin.

Should bevel be placed or not has always been controversial in composites.

Features favouring beveling are:

- The surface area is increased and the mechanical retention is improved.
- Marginal adaptability is improved.
- The prismless layer of surface enamel is removed.
- The ends of enamel rods are exposed for better etching.
- The color matching is improved since the transition from tooth to composite is gradual.

Features not in favour of beveling are:

- The orientation of prisms varies in different anatomical sites and beveling each site is difficult.
- Composite is liable to fracture, especially at stress-bearing areas.
- Optimum strength of restoration is achieved by providing butt joints.

The enamel margin in composite restoration should be beveled except at the occlusal surface. By beveling labial/lingual surfaces of anterior teeth, color matching is definitely improved. In case of cervical lesions, bevel should be placed only on occlusal/incisal wall and not on cervical wall. Similarly, bevel should be avoided in cervical/gingival cavosurface margins in proximal restorations.

Enameloplasty

It is defined as the procedure of reshaping or recontouring the enamel surface with suitable rotary cutting instruments to remove the shallow fissures by rounding or saucering the enamel so that the area becomes cleansable and finishable (Fig. 12.6A, B). Enameloplasty does not extend to the outline form.

Indications

- If one-third or less of enamel depth is involved, fissure may be removed by enameloplasty.
- Where the supplemental grooves extend till the cuspal inclines.
- Shallow fissure that approaches/crosses a lingual or facial ridge.

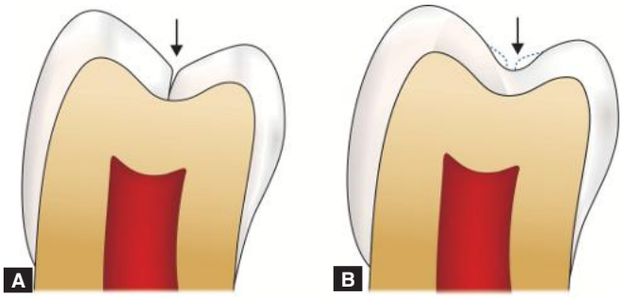


Fig. 12.6: Enameloplasty: **(A)** Shallow fissure (arrow); **(B)** Fissure round off (arrow)

Outline form: Key features

- Cavity margins are extended to sound tooth structure
- The unsupported enamel (weak enamel) margins are removed.
- Extend the margins to have sufficient access for cavity preparation, restoration and finishing/polishing procedures
- Location of the carious lesion
- Extent of the carious lesion
- Position of pits and fissures
- Proximity of the lesion to other defects in the enamel
- Relationship of adjacent and opposing teeth
- Type of restorative material to be used
- Functional requirements of the restoration
- Esthetic considerations
- Position and crest of healthy gingiva, age of the patient vis-a-vis the epithelial attachment and the scope for future gingival recession (the relationship of soft tissue).
- Dimensions of contact area (contact area is to be involved and later restored to same configuration)
- Extent of caries at and around the contact area
- Oral hygiene of the patient
- The forces of mastication, which the restoration will be withstanding
- Margins to be in self-cleansing areas; depending on restorative material
- Abnormal occlusal contacts and eating pattern of the patient; especially for planning cervical cavities.
- Contour of the buccal/lingual surfaces

2. RESISTANCE AND RETENTION FORMS

These two features are interrelated and interdependent; that is why considered simultaneously.

Resistance Form

Resistance form is defined as *"the shape and configuration of the cavity that best enables both the restoration and the tooth to withstand occlusal forces without fracture"*. The amount of remaining tooth structure and the choice of material influence the resistance form.

Features contributing to resistance form are:

- The cavity should be prepared as a box with a flat floor. Flat floor resists the occlusal stresses, being at right angle to such forces (Fig. 12.7A). Flat floor will help prevent the movement of the restoration; whereas, the rounded pulpal floor may allow rocking movement producing wedging force, which may result in shearing of tooth structure (Fig. 12.7B). Many a times, the flat floor is compromised at one or more than one place because of excavation of caries (Fig. 12.8A, B). Two/three flat seats may provide better resistance form to the prepared cavity (Fig. 12.8C).

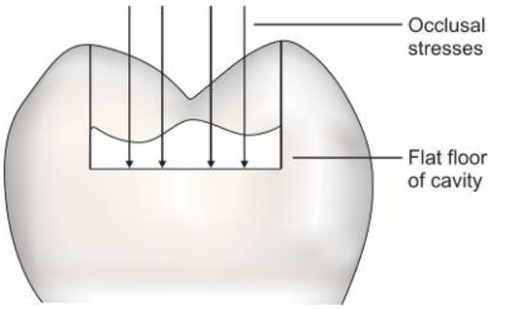


Fig. 12.7A: Flat floor resists occlusal stresses; provides resistance form

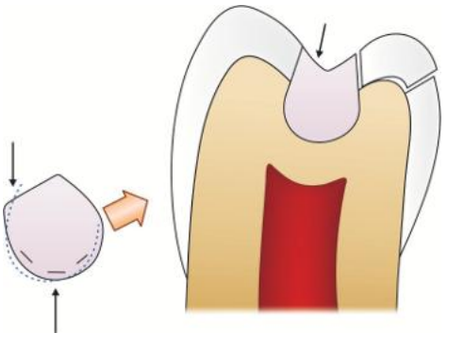


Fig. 12.7B: Cup-shaped cavity preparation (rounded pulp floor) may lead to tooth fracture

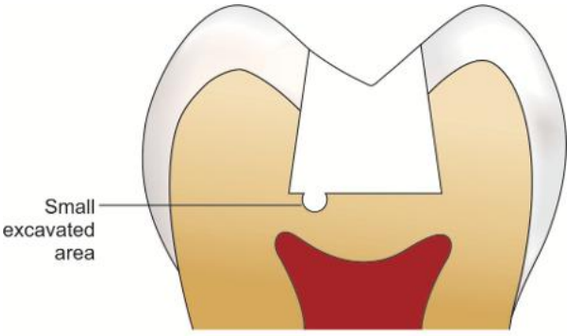


Fig. 12.8A: Flat floor compromised at one place

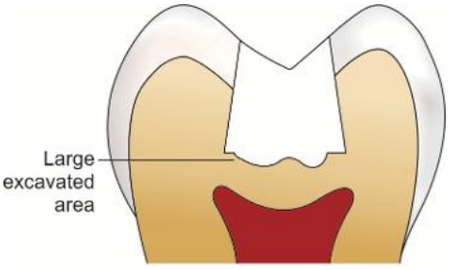


Fig. 12.8B: Flat floor compromised at two or more areas

- The extension of the external walls of the cavity preparation should be kept as small as possible. When a fissure is cut, the cavity margin should extend one-third distance from central groove to the cusp tip (Fig. 12.9).

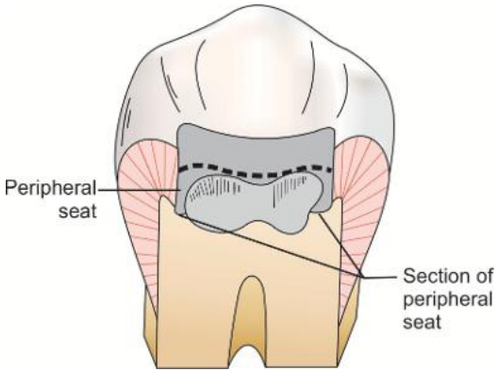


Fig. 12.8C: Resistance form improved with flat floor peripheral to excavated area (s)

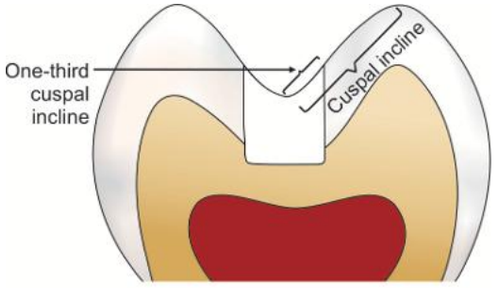


Fig. 12.9: Extension of cavity margin: One-third distance from the central groove to the cusp tip

- The restorative material should have enough thickness depending upon its respective compressive and tensile strengths to prevent its fracture under masticatory/functional load.
- Weakened cusps should be enveloped or included in cavity form.
- Rounding of line angles avoids stresses directly on to the tooth (Fig. 12.10), thereby resistance to fracture is increased [Sharp line angles and sharp proximal cavosurface margins lead to poor resistance form (Fig. 12.11A, B)].

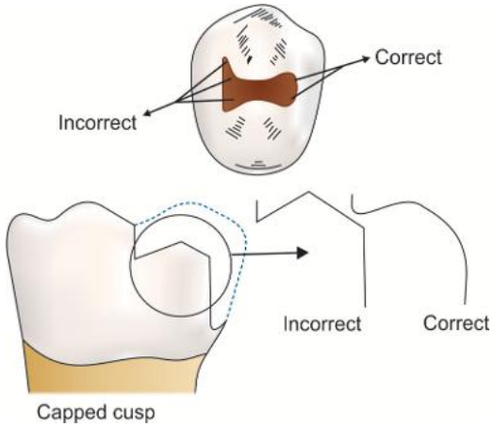


Fig. 12.10: Rounding of external and internal line angles

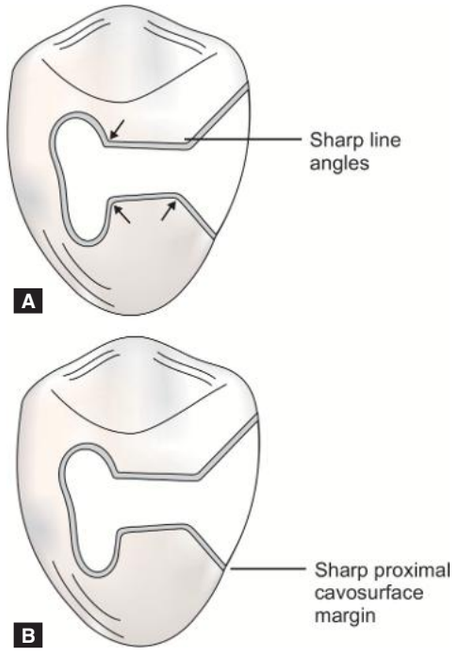


Fig. 12.11: **(A)** Sharp line angles (arrows); **(B)** Sharp proximal cavosurface margin

- Uniform depth: 1.5–2.0 mm overall/0.5–0.8 mm into dentin (Fig. 12.12). The minimal thickness of amalgam and cast gold to resist fracture is approximately 1.5 mm; little more depth is required for amalgam to achieve the requisite bulk. In composites and glass-ionomers, the depth is not the criteria for achieving resistance form. Porcelain also requires 2.0–2.5 mm depth for resistance form.
- Prefer restorative material having chemical bonding to tooth structure.
- Marginal ridges should be minimally encroached so as to preserve their strength (Fig. 12.13).
- All enamel walls must consist of full length enamel rods on sound dentin (Fig. 12.14A). The enamel

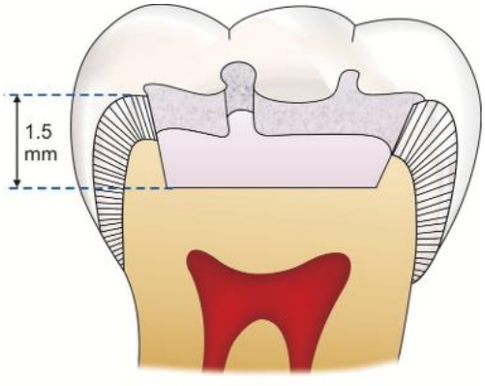


Fig. 12.12: Uniform depth of the pulpal floor

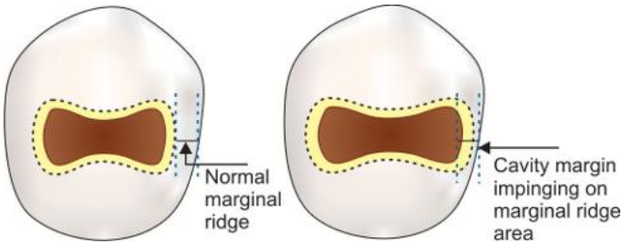


Fig. 12.13: Cavity margin in relation to marginal ridge area

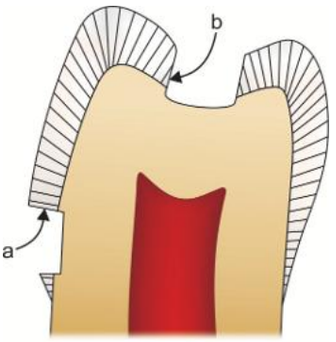


Fig. 12.14A: Enamel rods on sound dentin (a) full length; (b) short length

rods vary in orientation at different locations within the tooth. The angulation of enamel rods at occlusal third, middle third and cervical third is depicted in Fig. 12.14B.

- Cusp capping (covering of weak cusps to prevent fracture) is undertaken depending upon involvement of cusps.
  - If the intercuspal distance is one-third; no need of capping.
  - If the intercuspal distance is one-half; capping can be considered.
  - If the intercuspal distance is more than one-half; capping is mandatory (prior to planning capping, prefer reinforcing the lost dentin of weakened cusp with glass-ionomer cement) (Fig. 12.15).

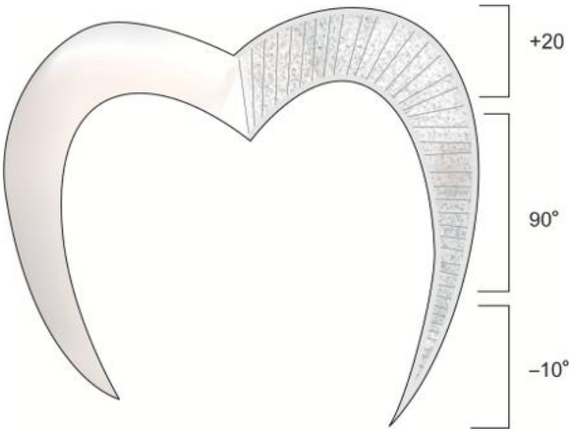


Fig. 12.14B: Orientation of enamel rods (longitudinal section)

Retention Form

'Retention form' is defined as *"the shape and configuration of the cavity that enables the restoration to be retained in that cavity under all types of tipping and tilting stresses".*

Features contributing to retention form are:

- Magnitude of the occlusal forces and the area of the restoration, which will be under the occlusal load.
- Configuration of the cavity; total surface area, (i.e. width and depth of the cavity)
- Available height of the cavity walls
- Amount of remaining dentin

Restorative Material and Retention Form

The retention form usually varies with type of the restorative material.

a. ***Retention form for silver amalgam:*** The retention form for silver amalgam is achieved by:
   - Converging the walls occlusally (approximately 2°), both for Class I and Class II cavities (Fig. 12.16A).
   - Slight undercuts can be given in the dentin (Fig. 12.16B).
   - Occlusal dovetail in Class II preparations (Fig. 12.16C).

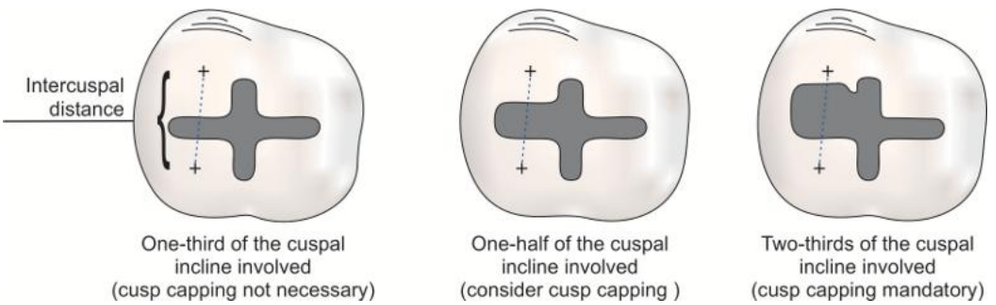


Fig. 12.15: Cusp capping

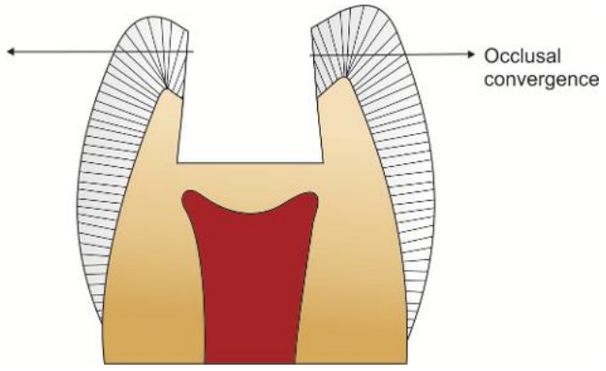


Fig. 12.16A: Occlusal convergence: Provide retention form

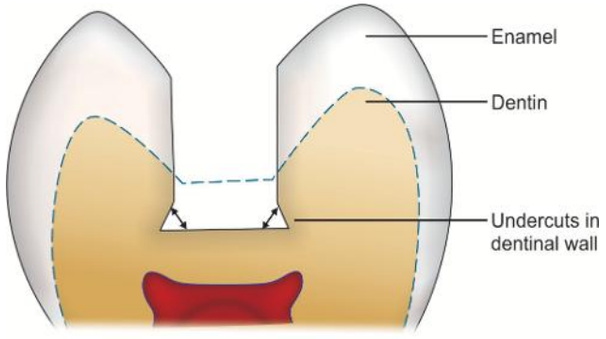


Fig. 12.16B: Undercuts in dentinal walls; provide retention form

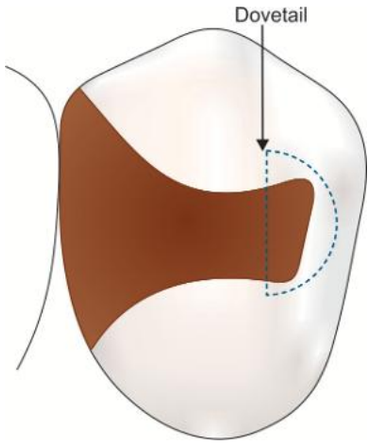


Fig. 12.16C: Occlusal dovetail in Class II preparation (maxillary first premolar)

- Micromechanical bonding to the tooth structure, decreasing microleakage.
- Cavosurface angle of 90°, i.e. butt joint [Marginal ridge should be conserved (Fig. 12.16D)].

b. ***Retention form for cast restorations:*** The retention form for cast restoration is achieved by:

- Parallelism of walls. A slight divergence (2°–5°) can be given for proper withdrawal of the pattern (Fig. 12.17). (In case the available height of the walls is less, the divergence should be kept minimum. At least one-half of the total wall should be kept parallel and the rest can be diverged.)
- Occlusal extension is mandatory even if no occlusal caries is present, since it prevents tilting. The outline form for the occlusal extension is same as for occlusal cavities (Fig. 12.18A, B).
- Reverse bevel at the gingival wall will prevent tipping movements.

c. ***Retention form for direct filling gold:*** In direct filling gold, the restoration is retained by the elastic compression developed in the dentin due to

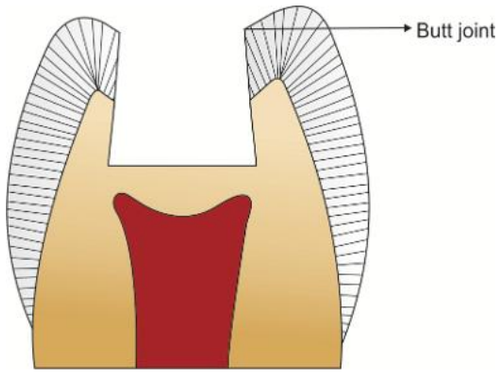


Fig. 12.16D: Butt joint at cavosurface margin: Maintain marginal ridge

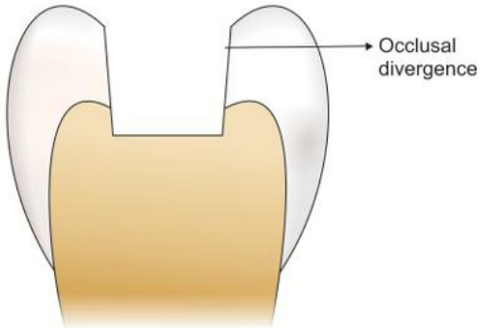


Fig. 12.17: Flat pulpal floor with occlusal divergence

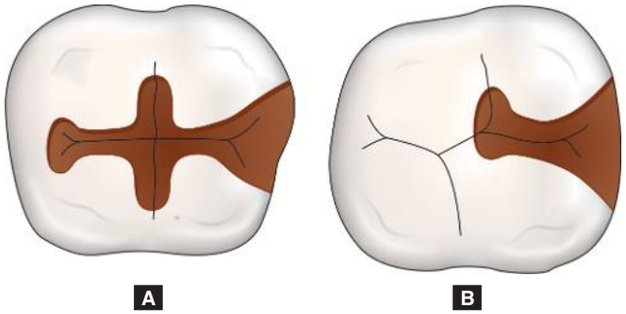


Fig. 12.18: Occlusal lock related to the morphology of: **(A)** Mandibular tooth; **(B)** Maxillary tooth

condensation of the foil; the starting points prepared in the cavity also aid in retention.

d. ***Retention form for composite restorations:*** In composite restorations, micromechanical bonding helps to achieve the retention form. The surface area of the cavity and the type of bonding agent, both aid in the retention form. It has been established that most of the bonding agents provide bond strength up to 22–24 MPa.

Secondary Retentive Devices

Many a times, the prepared cavity is not retentive even after providing the requisite retentive features. In such cases, 'secondary retentive devices' are used, which provide extra retention and may even reinforce the tooth structure.

The devices are as follows:

i. Slots
ii. Locks
iii. Grooves
iv. Skirts
v. Pins

i. ***Slots:*** Slots are given in dentin to increase the surface area of the preparation and also to have more convergent walls. These are 1.0–1.5 mm deep box type preparations and can be given in occlusal wall or gingival wall or both (Fig. 12.19A, B). Usually the slots are given all along the width of the occlusal/gingival wall. Each slot has four walls, which aid in further retention. Slots can be given with any type of restorative material.

ii. ***Locks:*** Locks are given in the proximal box of Class II cavity preparation and are indicated for silver amalgam restorations. Though not of much use, the locks are also tried with composites and glass-ionomers.

Locks are usually given in dentin, either in the walls of the proximal box or in the occlusal box at the line angles or axioproximal line angle. These

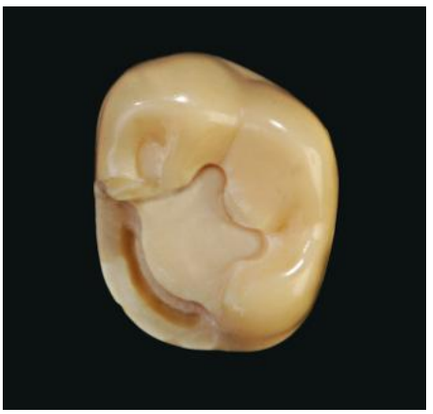


Fig. 12.19A: Slot preparation in maxillary second molar

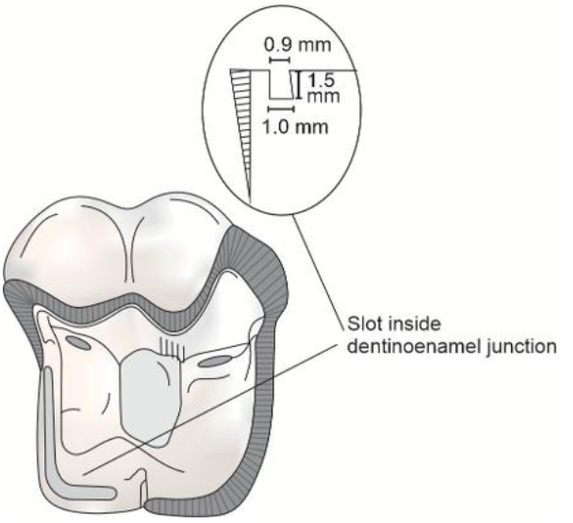


Fig. 12.19B: Slot preparation (diagrammatic)

are 0.2–0.3 mm wide and 0.5 mm deep into the dentin (Fig. 12.20A to C).

iii. ***Grooves:*** Grooves are indicated for cast restorations. These are prepared in the walls of the proximal box inside the dentinoenamel junction.

The depth of the groove should be more or at least equal to its width for providing proper retention. The grooves are placed at the axiobuccal and axiolingual line angles, putting more pressure on the buccal and the lingual walls rather than at the axial

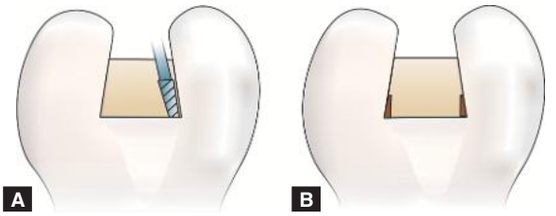


Fig. 12.20: Locks in the proximal box: **(A)** Fissure bur: preparing lock at the axio-buccal/lingual line angle; **(B)** Completed locks

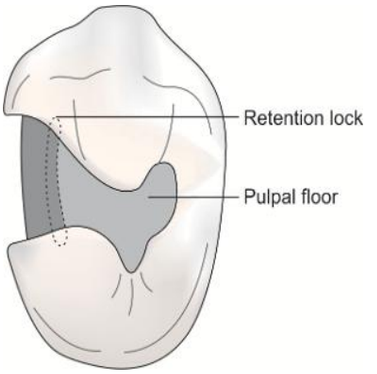


Fig. 12.20C: Prepared lock in premolar

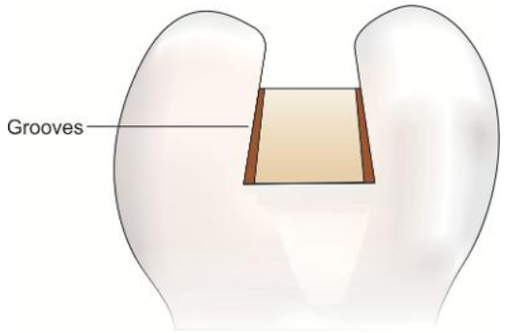


Fig. 12.21: Grooves in the proximal walls

wall (Fig. 12.21). The grooves are prepared parallel or slightly diverging occlusally for proper withdrawal of the pattern.

The grooves are also placed in veneer crown preparations (Fig. 12.22A to C).

iv. ***Skirts:*** Skirts, indicated in cast restorations, are the extensions of the proximal box at the line angles of the tooth or even away from it. The margins of the restoration are kept on healthy tooth structure and beveled. Such enveloping of the walls increases the total surface area of the restoration, thereby increasing the retention.

v. ***Pins:*** Pins, available in various sizes and shapes, provide extra retention when used with silver amalgam, composites and cast restorations. The use of pins is described in Chapter 17 (Complex Restorations).

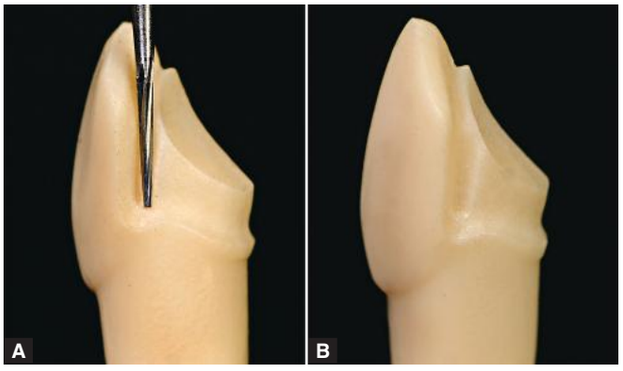


Fig. 12.22: Grooves in three-quarter crown (maxillary canine): **(A)** Preparation of proximal grooves; **(B)** 3/4th crown with groove (proximal view)



Fig. 12.22C: Half crown with grooves (buccal view)

Resistance and retention form: Key features

- Box form preparation with flat floor
- Depth of cavity / thickness of restorative material (1.5–2.0 mm depth sufficient for silver amalgam and cast restorations—slightly more for amalgam; 2.0– 2.5 mm for porcelain; depth not significant for composites and glass-ionomer cements)
- Occlusal convergence for silver amalgam and divergence for cast restorations
- Occlusal dovetail for better dissipating the occlusal forces
- Secondary retentive devices, viz. slots, skirts, grooves, locks, etc.

3. CONVENIENCE FORM

Convenience form implies that the operator must create sufficient access to the lesion to facilitate visibility and instrumentation. It is defined as '*the modification in shape of the cavity to have better access to the lesion, proper manipulation during restoration and finally for better finishing and polishing*'. Mostly, this form is required in proximal cavities, in both anterior and posterior teeth, i.e. Class II and Class III cavities.

Features influencing convenience form are:

- Location of the cavity margins (cavosurface margins be related to the restorative material)
- Accessibility for caries removal
- Ease of pulp protection procedures
- Accessibility for placing restorative material
  - Finishing of the cavity margins
  - Starting points, if any
  - Creation of other retentive means
- Type of the restorative material
- In Class II cavities, access is gained through the marginal ridge of the respective side.
- In Class III cavities, convenience form is achieved through lingual side, since labial enamel is to be preserved as far as possible.
- In Class I and Class V cavities, the convenience form is not difficult to achieve.
- Diverging of the walls of the cavity in cast restoration is a part of convenience form so as to have proper withdrawal of the pattern.

4. REMOVAL OF REMAINING CARIOUS DENTIN (FAULTY RESTORATIONS AND INFECTED DENTIN)

It is defined as *"the elimination of any infected carious tooth structure or faulty restoration left in the cavity preparation"*. After preparing the cavity and achieving adequate resistance and retention forms, the remaining portion is checked for any left-over caries. The site and location of such caries is important since it will dictate the future modification of the preparation. Caries left on the pulpal/axial wall is excavated slowly, maintaining 0.75–1.0 mm dentin covering the pulp. Caries will have to be excavated, even if it leads to exposure. Pulp capping is not required, if the remaining caries is hard, arrested or unaffected.

In no case, soft and infected caries should be left over the pulp. Such residual caries leads to pulpal inflammation and subsequently slow death of pulp. Chemical dyes, such as 0.5% basic fuchsin, may be helpful in differentiating the infected and the affected dentin. A few authors are also of the view that if some micro-organisms are left untreated and the cavity is filled, these become inactivated and get destroyed. However, it is advisable to remove all the infected dentin.

In case of children and young pulps, the remaining caries along the walls of the cavity should be excavated and the extent of such lesions is thoroughly checked. The undermining of enamel so produced can be treated in two ways:

i. The undermined area is filled with glass-ionomer cement making the configuration of the walls as required.
ii. Cuspal coverage can be considered in case of excess undermining.

The removal of remaining restorative material needs extra care, especially the tooth-colored restorative materials pose problem for their complete removal. Use of round burs at slow speed is recommended.

Pulp Protection

The pulpal injuries are usually caused by following factors:

- Heat generated by injudicious cutting
- Restorative materials with good thermal conductivity
- Chemical ingredients of the restorative materials
- Galvanic currents
- Ingress of micro-organisms through microleakage

The underlying pulp is to be protected, if the remaining dentin is less. It has been established that 2.0–2.5 mm remaining dentin is capable of countering external stimuli, therefore, procedure for pulp protection may not be required. In case the remaining dentin is less, pulp is to be protected using liners and bases depending upon the amount of dentin left and the restorative material to be used (Fig. 12.23A, B).

'Base' is the material, which is applied over the pulpal/axial wall and acts as a substitute for lost dentin. The thickness of the base depends upon the amount of dentin lost. The total bulk (dentin + base) should be at least 2.0 mm. The bases provide mechanical, chemical and thermal protection to the pulp. The materials like zinc phosphate cements, polycarboxylate cements, glass-ionomers, calcium hydroxide and zinc oxide eugenol are used as bases. In case of pulp exposure or very near to pulp exposure, a calcium hydroxide base is preferred below the usual cements (Fig. 12.23C, D). Recently introduced light cure calcium hydroxide can also be applied as a liner base.

The term 'liner' is used for those materials, which can be applied in a relatively thin film. The thickness of liners usually does not exceed 0.1 mm. Apart from providing thermal and chemical insulation, the liners fill the minor intricacies between the tooth and the restorative material. All the material used for bases can be used as liner; however, calcium hydroxide and zinc oxide eugenol are preferred.

Cavity Varnish

For shallow cavities, cavity varnish can be applied on the walls especially in silver amalgam and cast gold restorations. However, in composites and glass-ionomers, the application of cavity varnish is avoided.

Cavity varnish is a mixture of copal resin and organic solvents. On application, the solvents

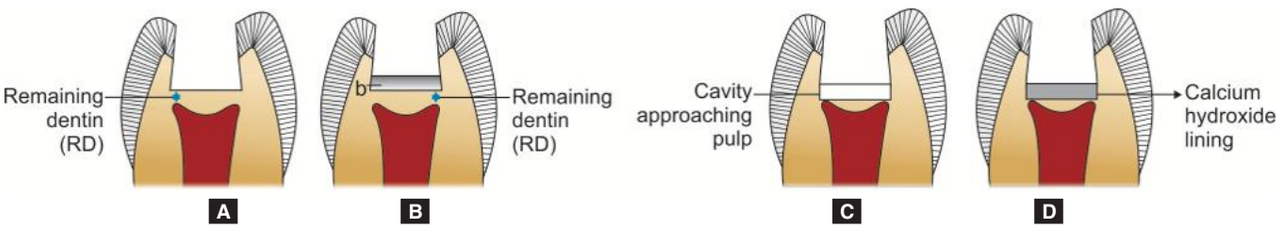


Fig. 12.23: Application of base: **(A)** Remaining dentin less than 1.0 to 1.5 mm; **(B)** A sub-base (b) of thickness 0.5 to 0.75 mm is applied; **(C)** Cavity approaching pulp; **(D)** Application of calcium hydroxide liner



Fig. 12.24: Cavity varnish

evaporate, leaving a resin layer over the dentin (Fig. 12.24). This seals the dentinal tubules, which leads to better marginal adaptability, especially with silver amalgam till the corrosion products fill the gap. Two coats of varnish are considered sufficient, since varnish is hydrophobic and with single application, 55% wetting is achieved (smear layer over the dentin has moisture). With the second application, 85–90% wetting is achieved, which is sufficient to provide the necessary sealing. Dentin-bonding agents are also recognized as the dentin-sealing agents and can be used as substitutes for cavity varnish.

Functions

- Prevents microleakage.
- Reduces postoperative sensitivity by not allowing ingress of fluid from the cavity margin.
- Prevents penetration of toxic materials into the dentin from the restorative material.
- Does not act as an insulator.

5. FINISHING THE ENAMEL WALLS AND MARGINS

The procedure is carried out to finish the cavosurface margins of the cavity walls so the restorative materials are adequately adapted to cavity walls; minimizing marginal leakage at the tooth–restoration interface.

This procedure creates a smooth junction between restoration and the tooth, and also provides maximum strength to both tooth and the restorative material, especially near the margins.

Features influencing cavosurface margins are:

i. Location of the margins vis-à-vis direction of the enamel rods.
ii. Type of the restorative material to be used.
iii. Degree of smoothness required (choice of restorative material can decide this):
   - For smooth walls—cast metal restorations
   - For rough walls—composite restorations, as they increase the surface area of restoration.
   - High speed to finish enamel.
   - Plain cut fissure burs produce the finest surface.
   - Hand instruments, viz. gingival margin trimmer and enamel hatchet to plain the enamel walls.
   - Not much smoothness is required in case of amalgam and gold foil.
iv. Location of the margins (whether under direct occlusal forces)

The usual anatomy of enamel rods is as follows:

- The rods extend full length from dentinoenamel junction to outer enamel surface.
- The enamel rods radiate from dentinoenamel junction to the external surface of the enamel. In axial sections, occlusally, the rods make a +20° to +30° angle with the long axis of the tooth. In the middle, they are perpendicular and in the gingival third they make an angle of –5° to –10° (Fig. 12.14B).
- The rods usually converge toward concave surfaces and diverge toward convex surfaces.

Finishing of enamel margins is carried out almost in every restorative material. In case of silver amalgam, butt end of the cavosurface margins is preferred because of poor edge strength of amalgam (Fig. 12.25A). The gingival wall is slightly beveled, thereby removing the unsupported enamel rods (Fig. 12.26). In gold foil restorations, an ultrashort bevel (40°) is created at the cavosurface margins. A short bevel is placed for cast gold restoration (Fig. 12.25B). Beveling, however, is avoided in porcelain. Beveling in composite is controversial; however, labial walls in Class III and occlusal walls in Class V cavities can be beveled for proper color matching.

6. TOILET OF THE CAVITY

The toilet of cavity implies removing all debris and cut dentinal chips from the walls of the cavity. Warm water can be utilized along with drying with oil-free air. A cotton swab can also be used to sweep out the debris. Mild acidic solutions have been tried but are not of much use (Fig. 12.27). Use of antiseptic solutions, alcohol, etc. should be avoided. The step of 'sterilization of cavity walls' could not gain much importance over the years, mostly because of advent of newer materials.

The overdrying of the cavity with compressed air is also avoided, especially in composite restorations; the bonding agents are hydrophilic and prefer wet dentin.

The 'conditioning' of the cavity floor and walls is carried out for glass-ionomer restorations. By using

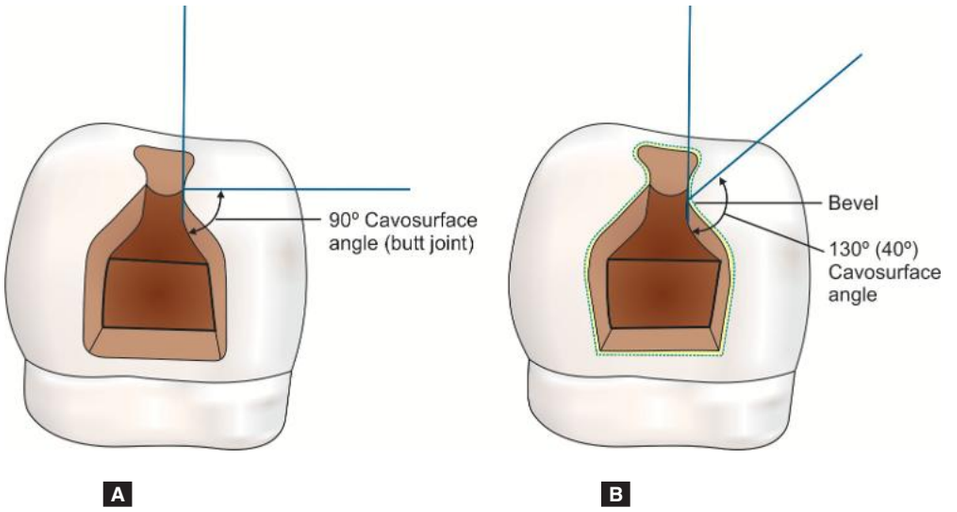


Fig. 12.25: Cavosurface margins: **(A)** 90° cavosurface angle (butt joint) for silver amalgam; **(B)** 130° (40°) cavosurface angle for cast gold

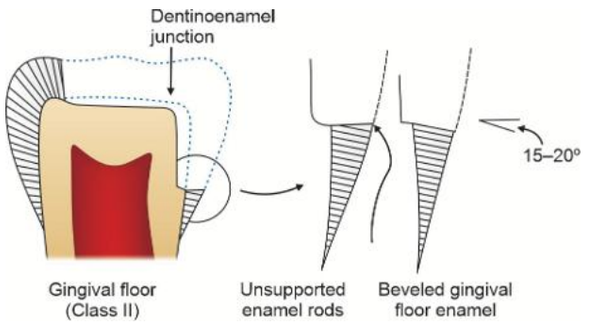


Fig. 12.26: Beveling the gingival floor in Class II cavity for silver amalgam



Fig. 12.27: Dehydrating and de-greasing agent for cavity cleansing

mild acids/etchant for 8–10 seconds, the surface calcium is reoriented, making it conducive for bonding with glass-ionomer. In case of 'etching', the inorganic salts are partially demineralized leading to enamel tags, which later help in mechanical union between the tooth and the restorative material. The smear layer is also removed or altered with this phenomenon (for details refer Chapter 18).

After cleaning the cavity before restoration, saliva/moisture should not be allowed to enter in the clean cavity.

Present Concept of Principles (Changing Paradigms)

After thorough academic discussion on the relevance of Black's principles, the consensus opinion as regard the changing concept of principles of cavity preparation is as follows:

a. Planning the cavity preparation
b. Initial cavity preparation
c. Final cavity preparation
d. Finishing the prepared cavity

a. ***Planning the cavity preparation:*** Before starting the cutting procedure, spend some time for planning the preparation. The final shape of the cavity is anticipated in mind keeping in view the following features:

i. Choice of the restorative material
ii. Extent of the carious lesion
iii. Occlusal contacts
iv. Shape of the contacts, contour and embrasures
v. Age of the patient
vi. Oral hygiene status and patient's awareness towards oral health

With the advancement in adhesive restorative materials coupled with patient's insistence for tooth-colored restorations, most of the clinicians are preferring composites as a material of choice for restoring both anterior and posterior teeth. In case of composites, the cavity preparation is kept

shallow and narrow since the tensile strength of these materials is good. However, in silver amalgam, the cavity is kept wider and deeper so as to have more bulk of the material. (The bulk provides proper resistance because of its poor tensile strength.)

With the use of cast restorations, the abnormal contacts, contours and occlusal disharmonies can be corrected. These restorations can also be planned for wider cavities involving more than two surfaces of the tooth. Cast restorations, especially the veneer crowns, need sufficient surface area and sufficient height of the cavity walls. In case sufficient height is not available, other treatment modalities, such as pin restorations, should be planned. The placement of margins in cast restorations depends upon the material to be used and the age, vis-à-vis the gingival condition of the patient.

In each case, patient's awareness towards oral health and oral hygiene is very important. In older age, with attrition and gingival recession, care should be taken to restore the tooth in such a way so as to utilize minimum chair time.

b. ***Initial cavity preparation:*** During initial cavity preparation, care is taken to remove all carious lesion, whether undermining enamel or not and whether near or over the pulp. The susceptible pits and fissures should be involved. Leaving susceptible pits and fissures, subsequently leads to secondary caries with all the restorative materials. The unsupported enamel should be removed.

c. ***Final cavity preparation:*** After initial cavity preparation, the cavity is checked thoroughly for any remaining caries, unsupported enamel and the depth of remaining dentin over the pulp. The final cavity preparation should be based on the following features:

i. Magnitude of occlusal forces
ii. Available surface area of the tooth
iii. Amount of remaining dentin left over the pulp
iv. Amount of remaining dentin under the cusps and ridges, and also whether these cusps and ridges are working or non-working
v. Extent of caries at the contact areas
vi. Extent of caries at the gingival areas
vii. Whether caries lesions are joining two or more than two surfaces
viii. Age of the patient, esthetic consciousness and extent of gingival recession
ix. Socio-economic status

The pattern of occlusal stress plays an important role in restorative dentistry. Every class of cavity, whether anterior or posterior, smaller or larger, will have to withstand the forces during mastication and during various movements of the mandible. Care should be taken to manipulate the working cusps and inclines properly. Strength of the working cusps is maintained by filling the undermining enamel with glass-ionomer cements and/or covering the cusps. Remember, a minimum of 3.0 mm of vertical height of walls is required to achieve resistance/retention form in cast restorations. In case such a height is less, the other sides of the tooth should be involved to counter the dislodging forces. Finally, if the surface for counter is not sufficient or total vertical height is much less than required, pin restorations can be considered.

Composites and glass-ionomers are not preferred in stress-bearing areas, though recent composites and pin-retained composites have shown some promising results. Composites are also being tried in wider cavities and in cases requiring cuspal coverage.

Cervical restorations should be given an extra time for final inspection because the pattern of occlusal forces in different individuals plays a vital role in retaining these restorations in place. The longevity of these restorations, vis-à-vis the marginal leakage and secondary caries also depend upon these stresses. Proper undercuts should be created and if need be, pins can be considered for achieving retention. The extension of lesion at the cementum should be dealt carefully as this area is more prone to leakage.

After giving final shape to the cavity, bases and liners should be placed to make the floor smooth and to protect the underlying pulp. With the advancement in composites and glass-ionomer cements, flat floors may not be the requisite, but protection of pulp is mandatory with every restorative material. Fast setting zinc oxide eugenol, zinc phosphate cement, glass-ionomer cements and light cure calcium hydroxide can be considered depending upon the need for substituting dentin.

d. ***Finishing the prepared cavity:*** The final cavity shall be thoroughly washed and cleaned before placing the restoration. Some debris might be clinging to the cavity walls. Washing with normal saline or warm water under pressure is helpful. Squeezing the cavity walls with cotton will further clean the cavity. The cavity should be dried with oil-free air and without any pressure. The enamel margins can be smoothened with hand-cutting instruments and the cavosurface margins are beveled depending upon the restorative material. The enamel rods at the gingival direction are to be dealt with

cautiously. In case the gingival wall is in cementum, it should not be beveled.

During the restorative procedures, the cavity is to be protected from saliva, moisture and other contaminants. The final phase of restoration is completed under strict isolation.

CONSERVATIVE CAVITY PREPARATION

It was as early as in 1920, Prime reported that the time has come to adapt the filling materials to the requirement of the tooth rather than force the tooth to meet the requirements of the filling material.

It is emphasized to keep the cavity cutting as minimal as possible; principle of conservative cavity preparation.

The features of clinical importance are:

i. The 'Outline form' of Black's cavity preparation, which includes the concept of 'extension for prevention' need attention. With the advent of adhesive restorative materials, this feature is no longer followed.

The width of occlusal outline is narrowed to 1.0 mm or even less. A wider occlusal cavity preparation as advocated by Black may weaken the tooth and lead to cuspal fracture. The reduction of intercuspal distance from 1/3rd to 1/4th resulted in greater strength in the prepared tooth.

ii. The extension of cavity into the dentinoenamel junction also needs evaluation.

The conservative cavity preparation is carried out only in enamel when the caries has not reached dentin. The usual thickness of enamel at the occlusal areas is 1.5 mm and the depth of the cavity can be kept 1.0 mm. The width of the cavity is approximately that of width of flat fissure bur (less than 1.0 mm). The direction of the enamel rods is such that the depth of cavity only in enamel can withstand the forces of mastication.

Conservative cavity preparation: Key points

1. Follow principles of minimally intervention dentistry.
2. Initially prefer preventive modalities; remineralization, enameloplasty, etc.
3. Manage occlusal anomalies prior to any preparation
4. Least amount of cutting (preparation) in any situation (Never remove more tooth structure than is absolutely necessary.)
5. Dental procedures be painless, less time consuming.
6. Use dental materials that conserve tooth structure.
7. Use long-lasting materials to minimize the need for future repair and replacement.
8. Prefer restorations that minimizes the number of appointments.
9. Prefer restorative materials that do not wear opposing teeth more than enamel.

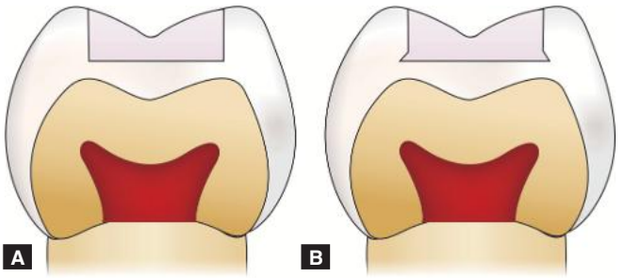


Fig. 12.28: Conservative cavity preparation only in enamel: **(A)** No undercut; **(B)** Undercut at the base

An ultra-small undercut is given at the base of the cavity for retention (Fig. 12.28A, B). A mini-condenser (0.7 mm diameter) is used to condense the silver amalgam into the cavity.

The cavity prepared only in enamel has the following advantages:

- Painless cavity preparation
- Strength of the tooth is maintained
- Chances of marginal breakdown are less
- Less time consuming and economical

iii. The feature 'extension of proximal walls into self-cleansing areas' is also questionable. The gingival sulcus is not bacteria free. Any restoration, which can be polished, is self-cleansable; therefore, extension of these walls into the embrasure is not required.

iv. Two parts of the cavity preparation, i.e. proximal half and the occlusal half; should be self-retentive. The proximal part can be retained by buccal and lingual retention locks, while the occlusal part by converging walls occlusally. Access to the proximal area is gained through the marginal ridge.

v. The 'tunnel restoration', though mainly used with glass-ionomer cement, has also been tried with silver amalgam restoration. It is used on a tooth with a carious lesion only on the proximal side and not involving the occlusal side. Similarly, ART technique is followed to minimize tooth cutting and restoring the tooth to its functions [refer Chapter 28 (Minimal Intervention Dentistry)].

Table 12.1 summarizes comparative features of conventional and conservative cavity preparation.

FORCES EXERTED DURING FUNCTIONAL OCCLUSION/MASTICATION

During movement of mandible and also during mastication, various forces are exerted on each tooth. Since the surfaces of the tooth are curved/inclined, these forces may be exerting at different directions other than vertical. The tooth, in turn, counteracts these forces with the support of periodontal membrane and alveolar bone.

Table 12.1: Comparative features of conventional and conservative cavity preparation

Conventional cavity	*Conservative cavity*
Class I Cavities	
• Extension for prevention (spread of future decay was in mind)	• Conservation; no extension for prevention (aim is conservation of tooth structure)
• Sweeping curves not indicated	• Sweeping curves preferred
• Wide and deep preparation	• Narrow and shallow preparation
• Occlusal convergence of mesial, distal, buccal and lingual walls	• Convergence only on buccal and lingual walls
• If cavities are on pits; pits are united in preparation	• If cavities are on pits; pits are prepared separately
Class II Cavities	
• Occlusal cutting follows principle of extension for prevention	• Follow conservative approach
• Proximal portion kept wide (bilateral truncated appearance)	• Proximal box kept narrow (small box or unilateral truncated appearance)
• Width of isthmus 1/3rd the intercuspal distance	• 1/4th the intercuspal distance
• No sweeping curves	• Sweeping curve is preferred ('S' shaped)
• Emphasis on wider isthmus	• Emphasis on narrow isthmus
• Reverse bevels placed in gingival seat	• No bevels in gingival seat
• Wide and deep restoration	• Smaller and narrow restoration
• Gingival seat is kept subgingival	• Gingival seat is kept supragingival
• Reverse curve emphasized	• Reverse curve not emphasized

If the surface is horizontal and perpendicular to the force of mastication, only vertical force would exert. But in curved or inclined plane/surface, other forces may be exerted along the long-axis of the tooth (Fig. 12.29 A, B).

This phenomenon can be better understood by studying the resolving of forces on inclined plane/surface. Resolving of forces refers to the process of finding two or more forces which, when combined, will produce a force with the same magnitude and direction as the original. The cuspal planes are taken as inclined planes.

When a force acts perpendicular to a fixed horizontal surface, the resolving force reacts perpendicular to the surface with an equal and opposite force. If the surface is inclined at an angle θ to the horizontal, it still reacts at right angle to the inclined surface with magnitude of F cosθ (Fig. 12.30).

Thus, the resolving force no longer opposes the applied force in direction nor is equal to its magnitude. Hence at that point of time, the resolving force is not in equilibrium when applied on inclined

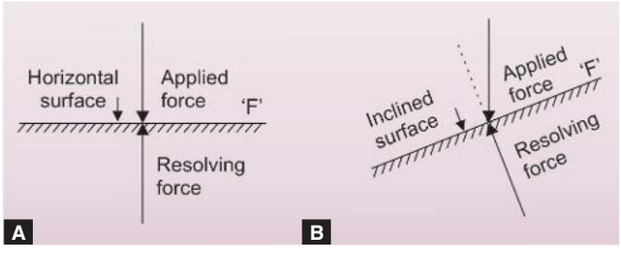


Fig. 12.29: Occlusion/mastication forces and its resolving force: **(A)** Horizontal surface; **(B)** Inclined surface

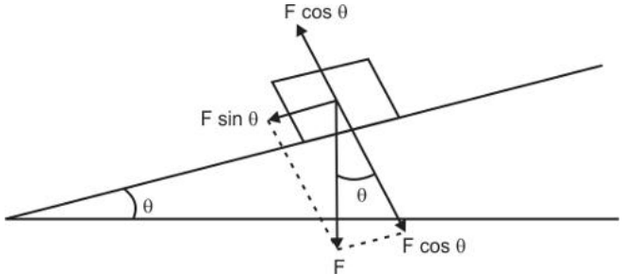


Fig. 12.30: Resolving force at incline plane

planes. But this equilibrium will be maintained, if the total applied force is resolved into horizontal (along the inclined plane) and vertical (right angle to inclined plane) directions.

Let us discuss how the forces act on cuspal incline planes (Fig. 12.31). AB is a line drawn at the contact point between two cusps and imply/make an incline plane at an angle 'θ' with the horizontal. F is the force of mastication and resolves into horizontal (F sinθ along the inclined plane/surface) and vertical components (F cosθ right angle to inclined plane/surface), thus maintains static equilibrium. As the angle decreases and becomes zero, then horizontal component, i.e. F sinθ will become zero as sin0 is zero and vertical component F cosθ will be equal to F as cos0 equal to one, which is similar to forces exerted on horizontal surface/plane.

The incline surfaces of one tooth contact the buccal/lingual cusps of the opposing tooth. This condition accounts for the proper balance in occlusion and in case the contact is not normal, it may account for displacements of the restoration/fracture of the teeth; the effect is termed as 'wedging effect'.

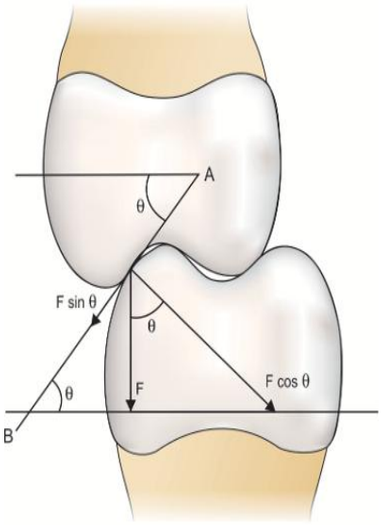


Fig. 12.31: Forces acting on cuspal inclines. F: Mastication force; Resolving forces; Horizontal: F sinθ and Vertical: F cosθ

The horizontal components of the applied force are responsible for this wedging effect. These horizontal components derived from the inclined surface/plane tend to push the buccal/ lingual cusps apart (Fig. 12.31).

At any given point of time, all forces at different directions will resolve itself and make a static equilibrium at teeth with a resultant force. *A resultant force (R) is the force (magnitude and direction) obtained when two or more forces are combined (i.e. added as vectors).*

Forces Acting on the Tooth

In centric occlusion, three forces, namely: F1, F2 and F3 are acting at three different points of contact: At a, b and c (Fig.12.32 A). At static equilibrium of teeth, point of contact a, b and c are in straight line.

Normally, force F1 and F2 are equal and acting in the same direction, but force F3 is acting in different direction. Since, F1 and F2 are equal and also acting in the same direction; therefore, their resultant force R_{12} (sum of F1 and F2) will act at a point of contact 'b' which is mid-point of 'a' and 'c'. If F1 and F2 are not equal then also their resultant force R_{12} will be sum of F1 and F2, but point of contact 'b' will be slightly shifted from the mid-point of contact 'a' and 'c'.

Now the two forces R_{12} (F1 + F2) and F3 are acting at a point of contact 'b' but in different direction, making an angle 'θ' with each other. During the static equilibrium of teeth, resultant of these forces R_{123} can be derived by "Vector addition with parallelogram method".

Let us understand the concept of parallelogram and vector:

Parallelogram is a simple quadrilateral (four sides) with two pairs of parallel sides. The opposite sides are equal length and opposite angles are equal measures (Fig. 12.32B).

Vector is a quantity that has both magnitude and direction. Vectors are frequently broken down into horizontal and vertical components (Fig.12.32C).

During static equilibrium of teeth, two forces, R_{12} and F3 are acting on a tooth at a point 'b' and making an angle 'θ' with each other. The resultant force R_{123} will be represented by the diagonal of the parallelogram. As per vector addition with parallelogram method, if two forces are acting simultaneously at a point by the adjacent sides of parallelogram, then the resultant of these forces will

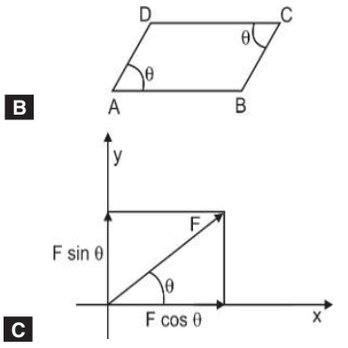


Fig. 12.32B and C: **(B)** Parallelogram; **(C)** Vector

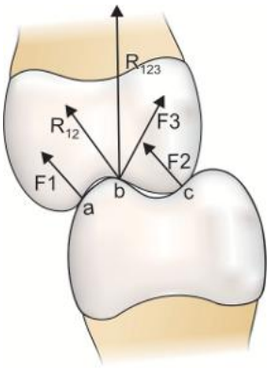


Fig. 12.32A: Forces acting in centric occlusion

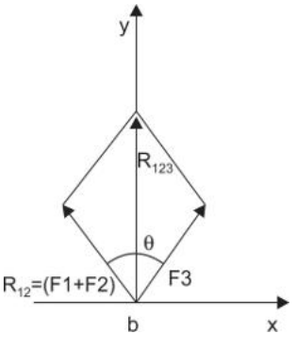


Fig. 12.32D: Parallelogram at point 'b'

be represented by the diagonal of the parallelogram (Fig. 12.32D).

Magnitude of resultant force R_{123} can be derived and will act in a vertical direction during static equilibrium of teeth.

Magnitude of R_{12} = F1+F2

Magnitude of R_{123} =

$$\sqrt{(R_{12})^2 + (F3)^2 + 2 \times R_{12} \times F3 \times \cos\theta}$$

During Chewing, (when mandible moves from centric occlusion to lateral position), the resultant of all three forces F1, F2 and F3 is not vertical but incline laterally. This is due to the fact that forces F1 and F2 are increased and F3 is decreased considerably. At one point of time, two forces R_{12} (sum of F1 and F2) and F3 are acting on a teeth at a point 'b' and making an angle 'θ' with each other. The resultant force R_{123} will be represented by the diagonal of the parallelogram. As per vector addition with parallelogram method, the resultant force R_{123} is shifted from vertical position towards lateral position and its magnitude can be found by the same equation as above. The resultant force R_{123} is inclined, therefore, it can be broken down into horizontal and vertical components (Fig. 12.32 E and F).

During chewing, (when mandible moves from lateral position to centric occlusion), the resultant of all three forces F1, F2 and F3 is also not vertical but incline medially. This is due to the same fact that forces F1 and F2 are increased and F3 is decreased considerably.

Similarly, two forces R_{12} (sum of F1 and F2) and F3 are acting on a tooth at a point 'b' and making an angle 'θ' with each other. The resultant force R_{123} will be represented by the diagonal of the parallelogram. As per vector addition with parallelogram method, the resultant force R_{123} is shifted from vertical position towards medially and its magnitude can be found by the same equation as

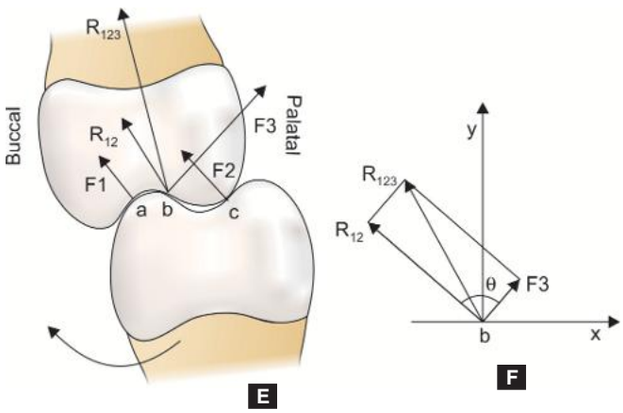


Fig. 12.32E and F: During chewing (when mandible moves from centric occlusion to lateral position)

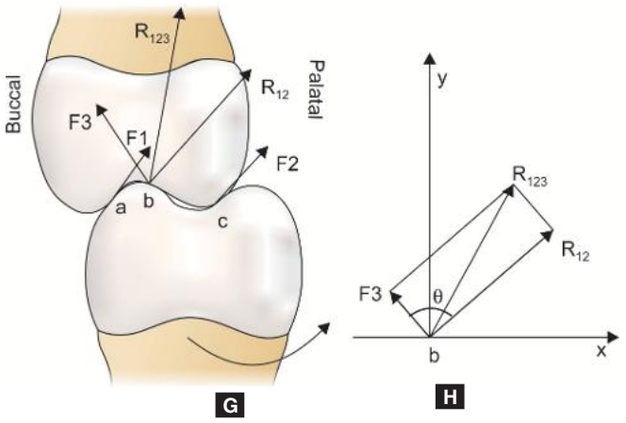


Fig. 12.32G and H: During chewing (when mandible moves from lateral position to centric occlusion)

above. The resultant force R_{123} is inclined, therefore, it can be broken down into horizontal and vertical components (Fig. 12.32 G and H).

The resultant force R_{123} is a thrust inclined palatally on the maxillary teeth and buccally on the mandibular teeth.

Marginal Ridge and Occlusal Forces

1. *Normal marginal ridge:* Forces 1 and 2 act on marginal ridges of teeth A and B, respectively. The horizontal component of 1, H_1 and the horizontal component of 2, H_2 counteracts each other. The vertical component V_1 and V_2 are resolved normally by the underlying tissues (Fig. 12.33).

2. *No marginal ridge:* In Fig. 12.34, tooth B has no marginal ridge. Forces 1 and 2 are acting on tooth A and B. The horizontal component of 2, H_2 is missing in the tooth B, because force 2 is mainly directly towards tooth A. Horizontal component H_2 is active on tooth A and may push tooth A away from tooth B. The vertical component V_1 and V_2 of forces 1 and 2 will help the food impact vertically. The vertical force V_2 will be more than required, there may occur slight tilting of the tooth B. This will further deteriorate the resolution of forces and lead to further food impaction (Fig. 12.34).

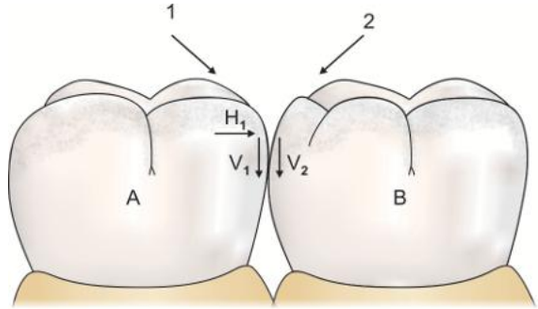


Fig. 12.33: Normal marginal ridge

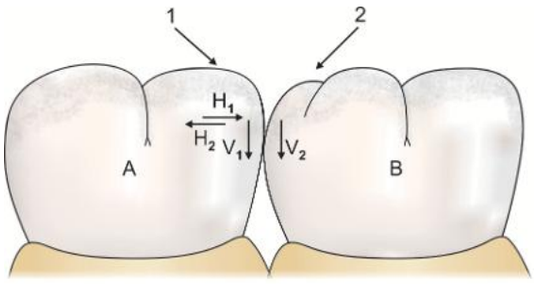


Fig. 12.34: No marginal ridge

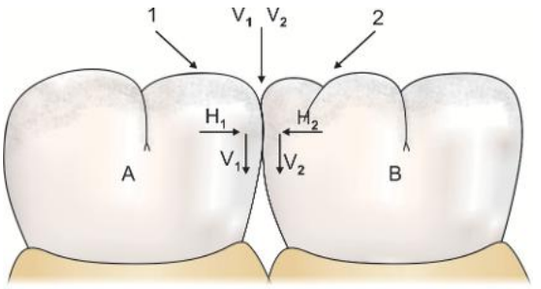


Fig. 12.36: Marginal ridge with no occlusal embrasure

3. ***Marginal ridge with wider occlusal embrasure:*** In spite of putting optimal pressure on marginal ridges of tooth A and B, the forces 1 and 2 act on adjacent teeth. The force 2 will put pressure on tooth A and force 1 will put pressure on tooth B. This will lead to drifting of both the teeth. The vertical component of forces will wedge the food in between the two teeth (Fig. 12.35).
   Similar effect is seen when one marginal ridge is higher than other.
4. ***Marginal ridge with no occlusal embrasure:*** The vertical component of forces 1 and 2 will be more concentrated than horizontal components. Though there will not be any vertical impaction of food, the continuous impact of higher concentration of vertical component of forces may lead to changes in alveolar bone after sometime (Fig. 12.36).

Vertical Loads and Distribution of Stresses

As the load is applied over the teeth, stresses are distributed both:

i. Parallel to the long axis, and
ii. Perpendicular to the long axis.

The stress distribution depends upon various factors; such as:

- If the cross-section of the area under stress is constant, stress distribution would be uniform.
- If there is variation in cross-section (such areas are normally termed as prisms), stress varies from point-to-point, being inversely proportional to the area under stress.

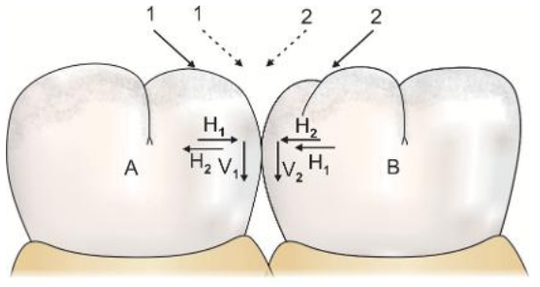


Fig. 12.35: Marginal ridge with wide occlusal embrasure

- If the cross-section area shows abrupt changes the stress would concentrate at that point.

The modulus of elasticity of the restorative material is an important property and should be taken care of. If a cavity is restored with gold inlay or porcelain, the modulus of elasticity varies between the tooth and the restorative material. The vertical force and the compression should be same for the restoration and the tooth; however, since gold/ porcelain is much stiffer, they will be highly stressed

$(S = \delta E)$

S (stress) = δ (unit strain) × E (modulus of elasticity)

When the force is applied perpendicular to the prism axis (area under stress, if not constant and varies in cross-section, is termed on 'prism'), the resultant resolution is known as 'beam'. Beam can be supported from both the ends (simple beam) and may be supported from one end (cantilever beam). The MOD preparations are the examples of simple beams; whereas, MO/DO preparations are the examples of cantilever beam. The retention of the restoration depends upon these beams, however, strength and the deflection of the material also play part.

Moment of force = Force × perpendicular distance

The bending moment is at the axiopulpal line angle, which tends to rotate the restoration out of the cavity.

The cavity dimensions are designed to counteract this moment. The total retentive force (R) is equal to F × L/l, where l is the depth of the axial wall. If we take depth of the gingival wall (d) into account, then R and d will be in the same direction, so their moment of force is zero. Therefore, the depth of the gingival wall does not take part in retention (Fig. 12.37).

In MOD preparation, the force (F) is divided equally on both the sides. The mesiodistal distance (L) is also divided into two. The moment of force at the midpoint is:

$$F/2 \times L/2 = FL/4$$

If this moment of force is divided into two (because it is actually acting on both the ends), then the moment of force is FL/8.

Since the beam forms a concave downward curvature between the load and the fixed end,

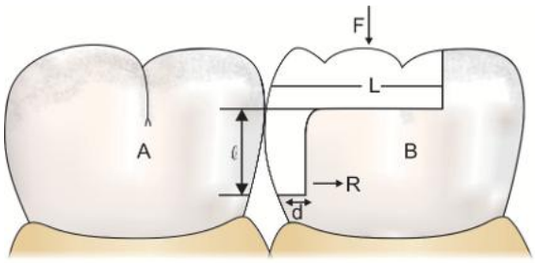


Fig. 12.37: Moment of force in MO/DO preparations

therefore, by sign convention, this end moment is taken as negative.

By equation $R \times l = FL/8$

So $R = FL/8l$

The negative sign is used only in vector form and in magnitude only positive sign is used.

In MOD preparations, it is presumed that the length of the axial wall (l) is kept equal on both the ends. If there is marked discrepancy between the two ends, the end result may not be the same as is described earlier. Therefore, preferably the length of the two axial walls should be the same (Fig. 12.38).

In cervical/gingival restorations, it has been established that certain forces act on the cervical region, which could destabilize the restoration and even lead to cracks at the cementoenamel junction.

In functional occlusion, the lingual slopes of buccal and palatal cusps of the maxillary teeth contact the buccal slopes of the buccal and lingual cusps of the mandibular teeth. The forces acting on inclined planes of the occluding cusps consequently lead to transverse stresses. These transverse stresses try to bend the tooth gingivoocclusally. Since the teeth are firmly held in alveolar socket, these rotations are minimum and counteracted.

In case the cavity is prepared on the cervical surfaces, depending upon the height of the axial wall, the deflective force is increased. If the restorative materials are not adhesive in nature, a gap can be created at the cervical surface of the restoration on buccal side and occlusal surface on the lingual side.

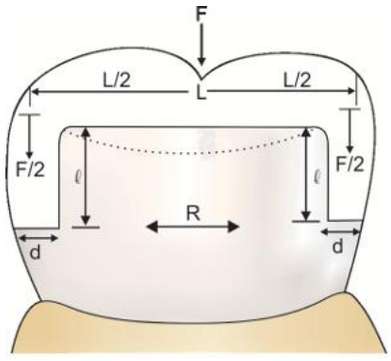


Fig. 12.38: Moment of force in MOD preparation

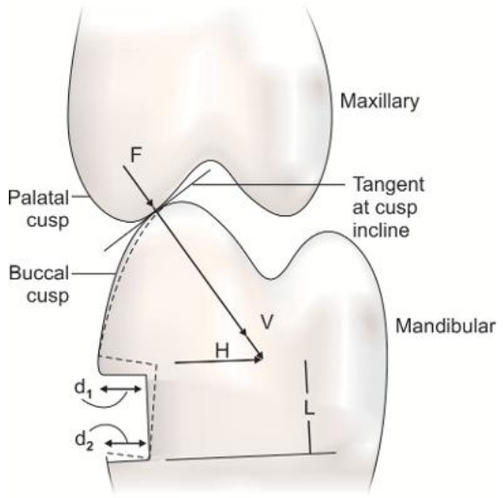


Fig. 12.39: Moment of force in cervical cavities

In Figure 12.39, buccal cusp of mandibular tooth and palatal cusp of maxillary tooth are in contact. Force (F) is applied at incline plane perpendicular to the tangent of the planes. The horizontal component (H) acts approximately at the center of the tooth. The vertical component (V) is constant. The deflection is mainly by the horizontal component, which depends upon the height of the axial wall (L) and depth of the occlusal (d_1) and the cervical walls (d_2).

Since, bending moments depend upon modulus of elasticity, a young tooth may deflect more because of its less modulus of elasticity.

Stress Distribution in Individual Restorations

Class I Restoration

Class I lesion can be restored with different materials. For restoration with amalgam, it is recommended to converge the side walls occlusally and to keep the floor flat. In case the floor is not flat, the forces will rotate the restoration on both the sides. And also, if the remaining dentin is less at the center, the forces might fracture the tooth.

The movement/rotation of the cast restoration is easy, if the pulpal floor is not flat. Since, the depth of the cavity is less and the walls are diverging occlusally, the chances of rotation are much more.

The rotational forces can be counteracted by adhesive materials, such as composite and glass-ionomer cement.

Class II Restoration

In Class II restorations, the stress, which tends to rotate the restoration (bending stress), mostly acts on marginal ridges.

Since, the bending stress acting at the axiopulpal line angle increases rapidly as its radius of curvature

decreases, the axiopulpal line angle should be rounded, so to increase the bulk of the material at this point. If the modulus of elasticity of the material differs with that of tooth, the stresses act as on a beam. In such cases, gingival lock is required to counter the rotation of the restoration.

In MOD restorations, centers of rotation are at the intersection of the neutral planes and not at the axiopulpal line angle. The bending of the occlusal portion is caused as in the other case, by the difference between the total masticatory force and the support given by the pulpal floor of the cavity. Gingival retention and roundening of the axiopulpal line angles are required as in MO/DO cavity.

In case, where the opposing cusps occlude in such a way that one contact point is on the proximocclusal restoration, while the other is on tooth structure, there is tendency to wedge the two apart. To prevent this wedging, the occlusal lock is used even though occlusal surface is not involved by caries.

Class III and Class IV Restorations

Since these lesions are not in direct contact with opposing teeth, only transverse stresses play part in dislodging/rotating the restoration.

In such restorations, there is tendency to rotate about an axis approximately parallel to the long axis of the tooth. As incisal retention cannot be made due to thin labiolingual size, so lingual lock is placed on lingual surface. It should be as close to the incisal edge as possible and still be in dentin to reduce the stress in this lingual lock.

In maxillary teeth, force of mastication has labial component, which provides the seating effect on the restoration. In case the labial enamel is not intact, the chances of dislodgement of the restoration will increase. In mandibular teeth, the component of the masticatory force is from the labial to the lingual, so chances of dislodgement of restoration are more.

Class V Restoration

The Class V restoration is not subjected to direct occlusal stresses; however, physical forces putting on occlusal surfaces could result in displacement of the restoration.

It is established that the buccal cusps of mandibular teeth and the palatal cusps of maxillary teeth contact each other during occlusion. The vertical stresses on the teeth lead to transverse stresses and this component of stresses tends to rotate the cervical restoration. The stresses increase with the cusp slopes of the maxillary teeth and the lingual inclines of the mandibular teeth. The mandibular teeth bend more than maxillary teeth. A gap is evident on the cervical/occlusal wall of the cavity and if the depth of these walls is less, the restoration may dislodge.

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